CITY OF MALABON UNIVERSITY
COLLEGE OF COMPUTER STUDIES

Integrated Event Registration, Payment, and Attendance Management System for City of Malabon University

A SIA 2 Project Presented to the
Faculty of College of Computer Studies
City of Malabon University
City of Malabon

In Partial Fulfillment of the Requirements for the Degree of
Bachelor of Science in Information Technology

By

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Executive Summary

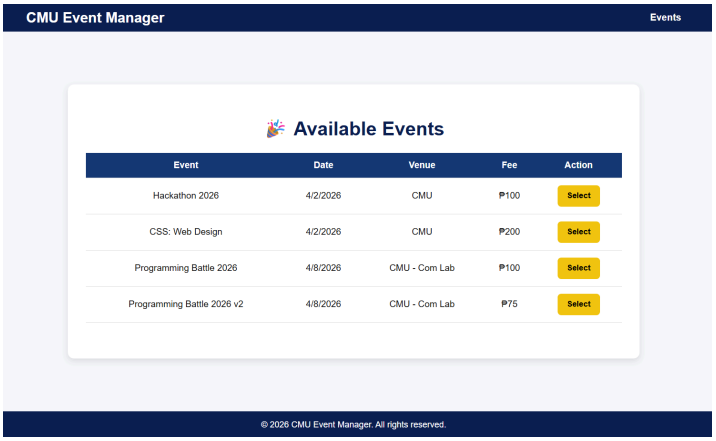
This project is a lightweight System Integration Architecture (SIA) that combines multiple tools into one automated event management system. It uses HTML, CSS, JS for frontend, Google Sheets as the database, Google Apps Script (GAS) for backend logic, n8n for workflow automation and integration, and GMail for notification. This tech stack is ideal because it connects different tools into one working system without needing complex backend infrastructure.

The problem with traditional event management is that registration, payment, and attendance are handled separately, leading to errors like duplicate entries, missed payments, and lack of centralized monitoring.

To solve this, the system integrates all processes into one flow. Google Sheets acts as the single source of truth with real-time data, GAS handles registration and data processing, and n8n automates email confirmations and workflows. This setup enables automatic participant ID generation, smooth payment tracking and verification, and real-time attendance validation during events.

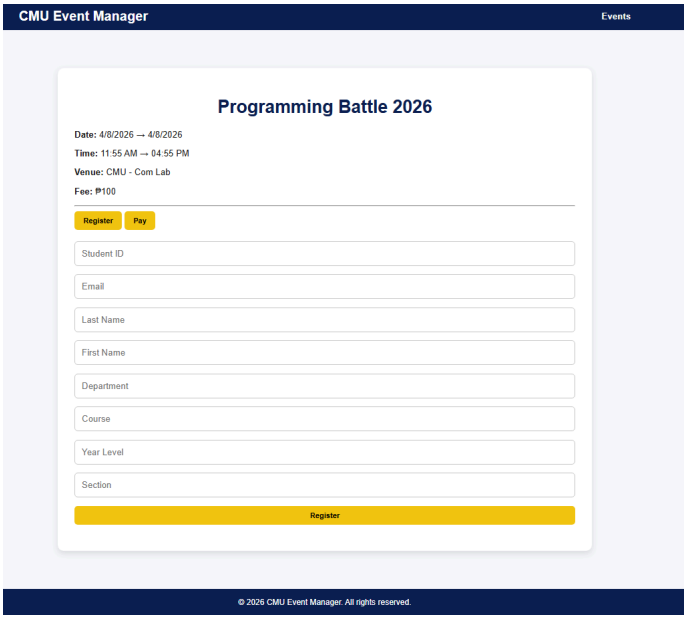
The system also defines clear admin roles: Level 4 (Master Admin) manages all events and system data, Level 3 handles specific events, attendees and IDs, Level 2 verifies payments and sets payment details, and Level 1 validates attendance at the event.

Attendees: Landing Page



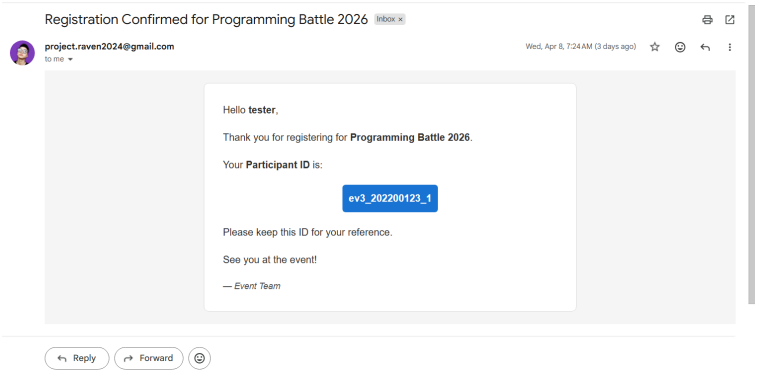
- Attendees can select the event they want to attend here
- Data from this comes from the Master Admin's Event Sheet

Attendees: Selected Event and Registration Page



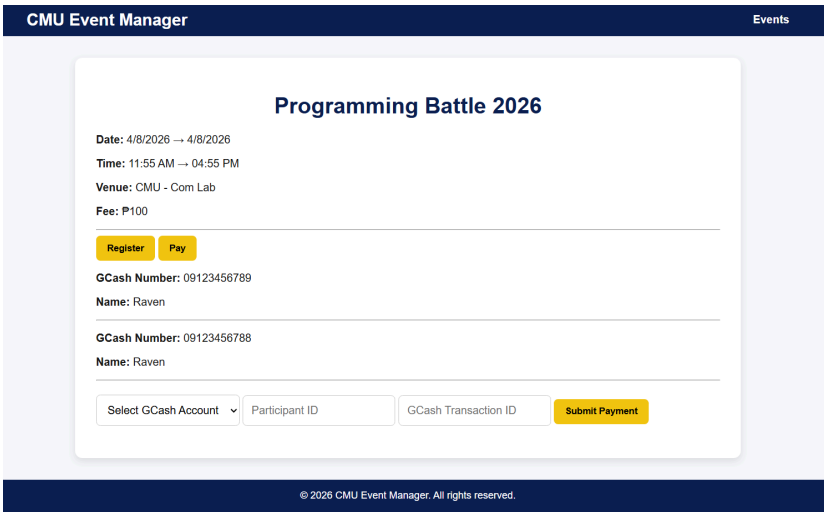
- Detailed view of the selected event and attendees' registration page

Attendees: Registration Email Confirmation



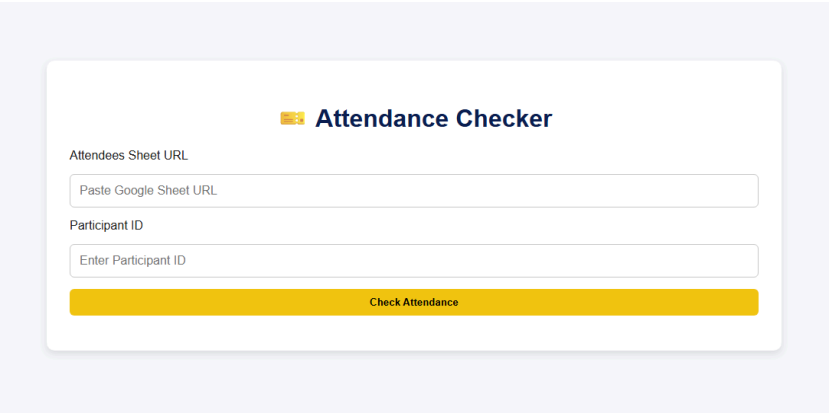
- Upon successful registration attendees will receive an email containing their Participant ID

Attendees: Event Payment Form



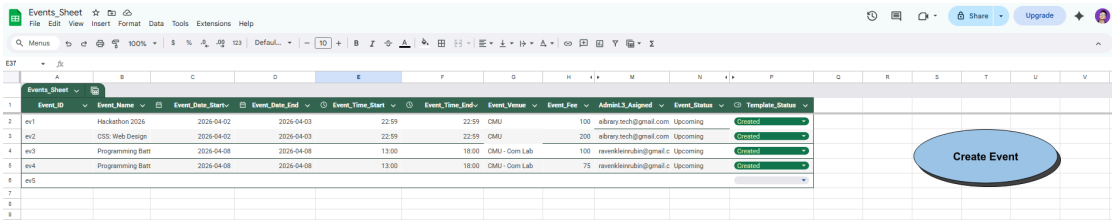
- Once attendees receive their Participant ID, they can now proceed on the Payment form in the same selected event page (If payment is required).

Attendees: Attendance Form (Manage by Admin Level 1)



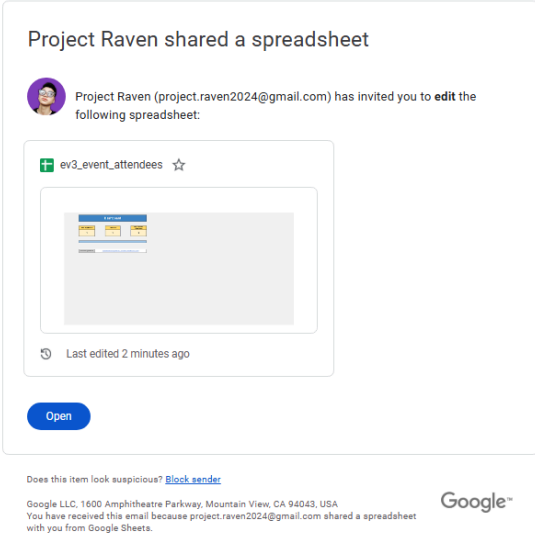
- For the event Gate access, admin (lvl1+) will check if the Participant ID exists and Payment is Paid in the Google Sheet database linked in the form; if true: event access granted.

Admin Level 4: Events Sheet (Master Admin)



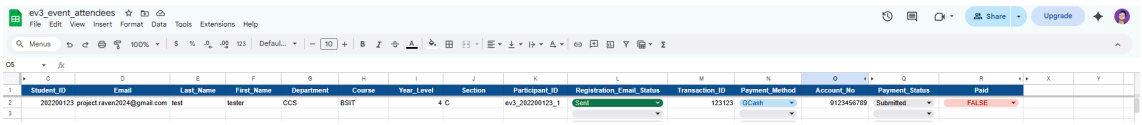
- This is the landing page UI of the Admin Level 4 or the Master Admin
- Data here will be shown in the “Attendees Landing Page”
- In this page, specific event sheets will be automatically created, granting admin level 3 access via Google Account.
- Admin use Google Sheet table for easy table management

Admin Level 3: Event sheet access invitation email

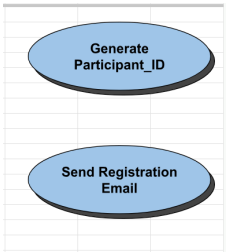


- Email invitation for admin level 3 to access their created event sheet.

Admin Level 3: Event Attendees Table

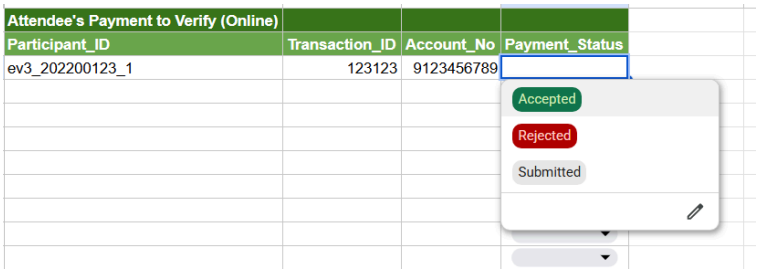


Student_ID	Email	Last_Name	First_Name	Department	Course	Year_Level	Section	Participant_ID	Registration_Email_Status	Transaction_ID	Payment_Method	Account_No	Payment_Status	Paid
202200123	project.raven2024@gmail.com	test	tester	CCS	BSIT	4	C	ev3_202200123_1	Sent	123123	GCash	9123456789	Submitted	FALSE



- Admin Level 3 can see all the event attendees here
- Generate Participant ID for new registration and send the email confirmation.

Admin Level 2: Attendee's Payment Verification

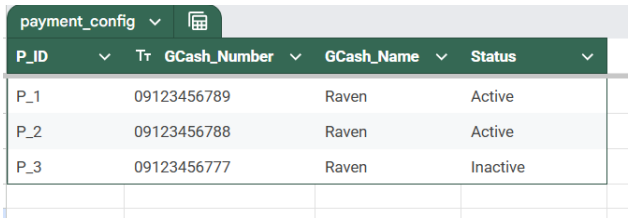


Attendee's Payment to Verify (Online)			
Participant_ID	Transaction_ID	Account_No	Payment_Status
ev3_202200123_1	123123	9123456789	

Accepted
Rejected
Submitted

- In the same event sheet file, "payment_verification" tab admin lvl 2 will verify if the Attendee's payment is valid.

Admin Level 2+: Payment Configuration

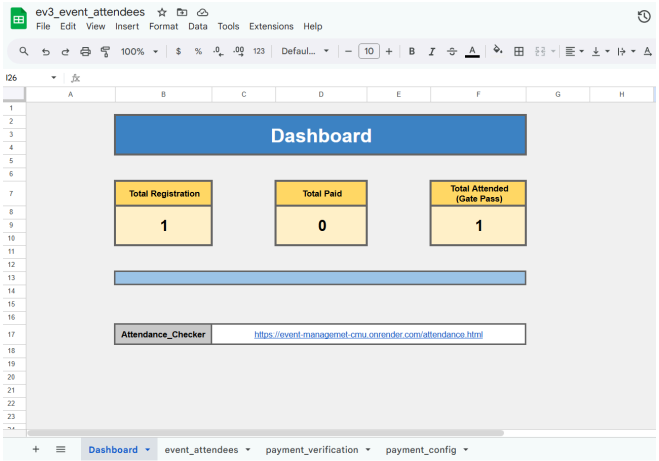


payment_config

P_ID	GCash_Number	GCash_Name	Status
P_1	09123456789	Raven	Active
P_2	09123456788	Raven	Active
P_3	09123456777	Raven	Inactive

- In the same event sheet file, "payment_config" tab admin lvl 2 + will set the GCash account on where the attendees will send their payment, this accounts will be shown in the "Attendees: Event Payment Form "

Admin: Dashboard



- Dashboard for the event sheet to track attendees data real - time; this is only the first version.